

Completing the Square

A general quadratic polynomial, $ax^2 + bx + c$, has the completed-square form $a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$. This formula always works (if you are using it, you should state the general formula for full credit! Just like with the quadratic formula). Some books or websites alternatively may write this slightly differently in "vertex form", $a(x - h)^2 + k$, where h and k constants with their own formulas. But what is the point of the completed-square form?

The point is that the standard completed-square form collapses down a quadratic polynomial with two x -terms into a single squared-binomial (involving x) and then some constants. And this squared-binomial is easy to isolate in many equations.

What if we want to complete-the-square without the use of this formula? Is there a general procedure?

e.g. 1) Write $3x^2 - 15x + 20$ in completed-square form.

Step 1) Look at the x^2 -coefficient and factor it out of the x^2 AND x terms (but not the constant term): $3x^2 - 15x + 20 = 3(x^2 - 5x) + 20$.

Step 2) Look at the x -coefficient inside the bracket and divide it by 2. In this case, that coefficient is -5 and dividing it by 2 gives us $-\frac{5}{2}$. Now this added to x is the binomial in our completed-square form i.e. $3x^2 - 15x + 20 = 3\left(x + \frac{-5}{2}\right)^2 + d$ for some unknown constant d .

Step 3) Solve for d by expanding and comparing constant terms (recall the "square-of-a-binomial" identity, $(a + b)^2 = a^2 + 2ab + b^2$):

$$\begin{aligned} 3x^2 - 15x + 20 &= 3\left(x + \frac{-5}{2}\right)^2 + d \\ 3x^2 - 15x + 20 &= 3\left(x^2 - 5x + \frac{25}{4}\right) + d \\ \cancel{3x^2} - \cancel{15x} + 20 &= \cancel{3x^2} - \cancel{15x} + \frac{75}{4} + d \\ 20 &= \frac{75}{4} + d \\ \Rightarrow d &= \frac{5}{4} \end{aligned}$$

Step 4) Write the completed-square form again with the knowledge of d . $3x^2 - 15x + 20$ in completed-square form is $3\left(x + \frac{-5}{2}\right)^2 + \frac{5}{4}$ or more simply $3\left(x - \frac{5}{2}\right)^2 + \frac{5}{4}$.

Now how could we use this completed-square form to find the roots of $3x^2 - 15x + 20$? Well, we want to solve:

$$3x^2 - 15x + 20 = 0$$

but we know that $3x^2 - 15x + 20 = 3\left(x - \frac{5}{2}\right)^2 + \frac{5}{4}$, so substituting in this completed-square expression:

$$\begin{aligned} 3\left(x - \frac{5}{2}\right)^2 + \frac{5}{4} &= 0 \\ 3\left(x - \frac{5}{2}\right)^2 &= -\frac{5}{4} \\ \left(x - \frac{5}{2}\right)^2 &= -\frac{5}{12} \\ x - \frac{5}{2} &= \pm\sqrt{-\frac{5}{12}} = \pm i\sqrt{\frac{5}{12}} \\ x &= \frac{5}{2} \pm i\sqrt{\frac{5}{12}} \end{aligned}$$

so the roots of $3x^2 - 15x + 20$ are $x = \frac{5}{2} \pm i\sqrt{\frac{5}{12}}$ or simplifying the radical $x = \frac{5}{2} \pm \frac{i}{2}\sqrt{\frac{5}{3}}$.

e.g. 2) Write $-x^2 - 6x + 1$ in completed-square form.

Step 1) Look at the x^2 -coefficient and factor it out of the x^2 AND x terms (but not the constant term): $-x^2 - 6x + 1 = -1(x^2 + 6x) + 1$.

Step 2) Look at the x -coefficient inside the bracket and divide it by 2. In this case, that coefficient is 6 and dividing it by 2 gives us 3. Now this added to x is the binomial in our completed-square form i.e. $-x^2 - 6x + 1 = -1(x + 3)^2 + d$ for some unknown constant d .

Step 3) Solve for d by expanding and comparing constant terms (recall the "square-of-a-binomial" identity, $(a + b)^2 = a^2 + 2ab + b^2$):

$$\begin{aligned} -x^2 - 6x + 1 &= -1(x + 3)^2 + d \\ -x^2 - 6x + 1 &= -1(x^2 + 6x + 9) + d \\ -\cancel{x^2} - \cancel{6x} + 1 &= -\cancel{x^2} - \cancel{6x} - 9 + d \\ 1 &= -9 + d \\ \Rightarrow d &= 10 \end{aligned}$$

Step 4) Write the completed-square form again with the knowledge of d . $-x^2 - 6x + 1$ in completed-square form is $-1(x+3)^2 + 10$.

Now how could we use this completed-square form to find the roots of $-x^2 - 6x + 1$? Well, we want to solve:

$$-x^2 - 6x + 1 = 0$$

but we know that $-x^2 - 6x + 1 = -1(x+3)^2 + 10$, so substituting in this completed-square expression:

$$-1(x+3)^2 + 10 = 0$$

$$-1(x+3)^2 = -10$$

$$(x+3)^2 = 10$$

$$x+3 = \pm\sqrt{10}$$

$$x = -3 \pm \sqrt{10}$$

so the roots of $-x^2 - 6x + 1$ are $x = -3 \pm \sqrt{10}$.